STAFF REVIEW · H446 093 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Memory location.

2. Content value/instruction bits.

3. Two operations compete for same limited resource.

4. Q becomes1, sampled at that active edge.

I Do — source worked steps

1. Explain the mechanism — Instruction and data transfers share memory/bus resources, so competing accesses can wait. — Describe the shared resource and its effect, not claim all Von Neumann machines are slow.

We Do — guided CHECK

1. Explain the mechanism
Harvard-style separation; fetch instruction and access data paths may operate in parallel.
It differs from a single shared path even when control logic is the same CPU.

You Do — source mark schemes

1. Von Neumann shared instruction/data storage/path [2], Harvard separate instruction/data memory/pathways [2].

2. Shared route [2], competing transfers [2], stalls or throughput effect [2].

3. Separate paths may reduce contention [2], real systems use modified Harvard caches [2], speed still depends on cache, clock and workload [2].

Exit Ticket — source answer

1. True. This is a common modified-Harvard arrangement.